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An Online Meta-Level Adaptive Design Framework with Targeted Learning Inference: Applications to Evaluating and Utilizing Surrogate Outcomes in Adaptive Designs

Wenxin Zhang^{1,#}, Aaron Hudson^{2,3}, Maya Petersen¹, Mark van der Laan^{1,*}

¹Division of Biostatistics, University of California, Berkeley

²Vaccine and Infectious Disease Division, Fred Hutchinson Cancer Center

³Department of Biostatistics, University of Washington

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#wenxin_zhang@berkeley.edu

Abstract

Adaptive designs are increasingly used in clinical trials and online experiments to improve participant outcomes by dynamically updating treatment allocation as data accumulate. In practice, experimenters often consider multiple candidate designs, each with distinct trade-offs. However, typically only one design is implemented at a time, leaving benefits and costs of alternative designs unobserved and unquantified. To address this, we propose a novel meta-level adaptive design framework that enables real-time, data-driven evaluation and selection among candidate adaptive designs. Specifically, we define a new class of causal estimands to evaluate adaptive designs and propose Targeted Maximum Likelihood Estimators for these estimands. These estimators are asymptotically normal while accommodating dependence in adaptive-design data without parametric assumptions, enabling online selection among candidate designs. We further apply this framework to a motivating example where multiple surrogates of a long-term outcome are considered for updating randomization probabilities in adaptive experiments. Unlike existing surrogate evaluation methods, our approach comprehensively quantifies surrogates' utility to accelerate detection of heterogeneous treatment effects, expedite updates to treatment randomization, and improve participant outcomes, facilitating dynamic selection among surrogate-guided designs. Overall, our framework provides a unified approach for evaluating opportunities and costs of various adaptive designs and guiding real-time decision-making in adaptive experiments.

Keywords: Adaptive Experimental Design, Causal Inference, Targeted Maximum Likelihood Estimation, Surrogacy

1 Introduction

1.1 Overview

Adaptive experimental designs are gaining popularity in clinical trials and online experiments (Robertson et al., 2023; Larsen et al., 2024). Unlike traditional static experimental designs, adaptive designs allow continuous updates to treatment randomization probabilities and/or other experimental features based on accumulated data during the experiment, supporting a range of experimental objectives. A key advantage of adaptive designs is their ability to increase the probability of providing participants with their optimal treatment, particularly when this optimal treatment varies between individuals, thereby improving their outcomes.

However, in practice, experimenters often face the challenge of choosing among multiple candidate adaptive design configurations, each with its own trade-offs. A common and compelling example is an adaptive experiment with a longitudinal data structure, in which new participants are continuously enrolled and existing participants generate multiple short-term outcomes that unfold progressively over different follow-up times, while the primary outcome requires long-term follow-up. In such settings, surrogate outcomes, either the early proxies of the primary outcome or other relevant metrics requiring shorter follow-ups, have the potential to quickly identify more effective personalized treatments, inform timely adaptations of treatment randomization probabilities with sufficient data, and thereby enhance participants' welfare. However, when multiple candidate surrogates are available, selecting among them involves complex trade-offs. Early surrogates provide more abundant and timely data for rapid adaptation, but may be less accurate in detecting treatment effects relevant to the primary outcome. Later surrogates may offer more accurate guidance for optimal treatment, but their delayed availability and limited sample size reduce their impact on early adaptation and participant benefit. In addition, the sensitivity with which a candidate surrogate captures heterogeneous treatment effects is a key driver of its utility: a surrogate that provides stronger signals for distinguishing which individuals benefit from which treatments enables more timely and targeted adaptation toward personalized treatment. Despite all these important factors, in practice, only one candidate adaptation can be implemented at each time point of the experiment, which inevitably leaves the gains and losses of unchosen designs unobserved and unquantified, and further complicates the challenge of navigating these trade-offs in a real-time experiment, underscoring the need for a rigorous statistical framework to overcome these difficulties.

To address these challenges, we propose an online meta-level adaptive design framework with targeted learning inference: a general and data-driven adaptive design system that continuously evaluates and dynamically updates its choice among different candidate designs in real time during ongoing adaptive experiments. We refer to this unified framework as TMLE-OSLAD, a Targeted Maximum Likelihood Estimation-based Online Super Learner Adaptive Design, which itself constitutes a meta-level adaptive design. This framework comprises three key components, each making a distinct methodological contribution.

First, the proposed framework introduces a novel counterfactual perspective for adaptive experiments by asking: How might participants' outcomes have differed under alternative adaptive designs? To formalize this question, we define a new class of target estimands: the mean counterfactual primary outcome that would have been observed had each participant's treatment, beginning at their enrollment, been assigned according to treatment randomization functions specified by the candidate adaptive design.

Second, to enable valid inference from the non-i.i.d. data generated by adaptive experiments, we establish a Targeted Maximum Likelihood Estimation (TMLE) approach that yields a consistent and asymptotically normal estimator without parametric assumptions, building upon martingale central limit theorems and equicontinuity results via maximal inequalities for martingale processes. We further leverage recent advances in higher-order Highly Adaptive Lasso (van der Laan, 2023) to provide equicontinuity conditions. These developments enable robust inference on the proposed causal estimand—the expected outcome under a candidate counterfactual adaptive design. At the end of the experiment, we also provide TMLEs for estimating other causal estimands of interest, such as the average treatment effect (ATE) and the mean outcome under optimal dynamic treatment regimes (rules) (Murphy, 2003; Robins, 2004; van der Laan and Luedtke, 2015).

Third, we develop a TMLE-based Online Super Learner Adaptive Design, which continuously updates estimates of the proposed estimand to evaluate and compare a library of candidate adaptive designs in real time, selects the data-driven best-performing design based on TMLE estimation and inference, dynamically adapts treatment randomization probabilities according to the chosen design, and continuously monitors and reevaluates the set of candidate designs throughout the experiment.

Returning to the motivating example of adaptive experiments involving surrogates, TMLE-OSLAD demonstrates an additional methodological advantage: it provides an adaptive design perspective for rigorously quantifying the value of information flow as it unfolds over time and is continually leveraged to guide sequential decision-making. Specifically, TMLE-OSLAD introduces, to the best of our knowledge, for the first time, a comprehensive causal estimand to evaluate the evolving surrogate utility in such dynamic settings with statistical inference. In this context, the target estimand is defined as the mean counterfactual primary outcome under a candidate surrogate-guided adaptive design, where each participant's treatment randomization probability at enrollment is determined by the estimated conditional average treatment effect function of the surrogate outcome and adaptively tilted toward the estimated optimal personalized treatment with the objective of optimizing the surrogate outcome. This metric reflects all critical factors of the utility of a surrogate in accelerating participant benefit through sequential adaptive design, including the timeliness of surrogate availability for enabling early adaptation, its sensitivity to signal heterogeneous treatment effects, and its predictive accuracy for identifying the optimal personalized treatments for the primary outcome. This is distinct from existing methods that focus on evaluating static statistical associations or causal relationships between surrogate and primary outcomes rather than adopting a dynamic, sequential decision-making perspective.

Furthermore, TMLE-OSLAD enables a principled, data-driven implementation of adaptive experiments that facilitates real-time evaluation and selection among different surrogate-guided adaptive designs to improve the primary outcome.

In summary, TMLE-OSLAD provides a comprehensive meta-level adaptive design framework for monitoring and quantifying opportunities and costs across a library of candidate adaptive experimental strategies. These strategies may differ in the quantities that guide adaptation, the statistical methods used to estimate them, or the rules governing design updates. The framework dynamically selects among these strategies as data accumulate, empowering data-driven, statistical-inference-based decision-making in diverse adaptive experimental settings.

1.2 Related Work

Adaptive experimental designs. Adaptive design is an active research area in statistics. In clinical trials, adaptive design methods are useful due to their flexibility and efficiency (Chow and Chang, 2008). These designs enable adaptive modifications to key aspects such as eligibility criteria, randomization probabilities, and other design features based on sequentially accrued data. Among these, Response-Adaptive Randomization (RAR) updates treatment probabilities using treatment and outcome data of previously enrolled subjects to achieve estimation efficiency or enhance participant welfare by reducing assignment to inferior treatments (Hu and Rosenberger, 2006; Robertson et al., 2023). Covariate-Adjusted Response-Adaptive (CARA) designs extend RAR by incorporating baseline covariates into randomization, enabling more personalized allocation and improved efficiency (Rosenberger et al., 2001; Hu and Rosenberger, 2006; Zhang et al., 2007; van der Laan, 2008; Rosenberger and Sverdlov, 2008; Atkinson et al., 2011; Cheung et al., 2014). One can also guide CARA for multiple objectives, e.g., by integrating both efficiency and ethics considerations in adaptation procedures (Hu et al., 2015), and by both balancing prognostic covariates and facilitating adaptation procedures using predictive covariates in CARA designs (Zhao et al., 2022). The origins of RAR and CARA designs also trace back to Thompson (1933) and Robbins (1952) and connect to multi-armed and contextual bandit literature (Bubeck et al., 2012; Lattimore and Szepesvári, 2020; Simchi-Levi and Wang, 2023). Our work differs by considering settings where multiple candidate adaptive designs are available to choose from and providing a principled way to navigate their trade-offs and select from these designs by TMLE-OSLAD framework, with applications to CARA designs with surrogate outcomes.

Surrogate outcomes. Surrogate outcomes are often considered as a substitute for the primary outcome when the latter requires a long follow-up time or is expensive to measure. In clinical trials, using surrogate outcomes can result in earlier detection of treatment effects, accelerated drug approvals, and reduced costs (Buyse et al., 2000). Examples include vaccine-induced immune responses as endpoints for HIV protection (Gilbert and Hudgens, 2008) and predictive short-term metrics to facilitate digital experiments (Duan et al., 2021; Richardson et al., 2023). The use of surrogates when primary outcomes are delayed has also been discussed in the adaptive design literature (Robertson et al., 2023). Examples include using short-term responses to guide treatment allocation for outpatients with depressive disorder (Tamura et al., 1994) and for participants in stroke trials (Nowacki et al., 2017), Bayesian modeling of surrogate–primary relationships to support RAR designs (Huang et al., 2009), and contextual bandits that handle

delayed rewards by running pre-trained Bayesian reward models (McDonald et al., 2023) or imputing delayed outcomes using surrogates constructed from external data under surrogacy assumptions (Yang et al., 2024). Relatedly, Gao et al. (2024) analyzes RAR designs driven by an informative surrogate and provides consistent, asymptotically normal statistical inference for the primary outcome under exponential-family models. While these approaches implement RAR or CARA procedures under a specific surrogate-guided design, a unified framework is lacking to dynamically re-evaluate and select among multiple surrogates to guide adaptive experiments.

Another related research area is surrogate evaluation. Many statistical frameworks have been developed to evaluate surrogates (Prentice, 1989; Freedman et al., 1992; Robins and Greenland, 1992; Daniels and Hughes, 1997; Lin et al., 1997; Gilbert and Hudgens, 2008) and reviewed comprehensively (Buyse et al., 2000; VanderWeele, 2013; Weir and Taylor, 2022; Elliott, 2023). However, most do not account for the potential benefits and risks of using surrogate outcomes in adaptive experiments that aim to learn and respond to treatment effect heterogeneity. Navigating these trade-offs requires both new approaches for quantifying the benefit of candidate surrogates, and novel designs and estimators that make possible online evaluation of these benefits and dynamic surrogate selection in response during the course of a study, which is addressed by our general TMLE-OSLAD framework.

Inference based on adaptive design data. The statistical inference procedure in TMLE-OSLAD is related to the problem of how to provide valid inference for adaptively collected data in the adaptive design literature. For example, asymptotic inference under specified parametric models has been established for data generated by CARA designs (Zhang et al., 2007; Zhu, 2015). Within the Targeted Maximum Likelihood Estimation framework (van der Laan and Rubin, 2006; van der Laan and Rose, 2011, 2018), van der Laan (2008) developed TMLE tailored to adaptive designs without relying on specific parametric assumptions; subsequent work employed TMLE to estimate risk difference and log-relative risk (Chambaz and van der Laan, 2014) and the mean outcome under optimal dynamic treatment rules (Chambaz et al., 2017) in CARA designs. Another line of work develops inference methods for adaptively collected data based on variance-stabilizing weighting techniques introduced by Luedtke and van der Laan (2016a) in non-adaptive settings and extended to treatment effect estimation or policy evaluation in multi-armed and contextual bandits (Hadad et al., 2021; Bibaut et al., 2021; Zhan et al., 2021), as well as to M-estimation of parametric models using adaptively collected data (Zhang et al., 2021). Our proposed target estimand, the mean outcome under counterfactual adaptive designs, is distinct from these methods and related to causal inference for time series (van der Laan and Malenica, 2018; Malenica et al., 2021, 2024). We develop a TMLE for this estimand, and establish asymptotic normality by martingale theories and equicontinuity results. We also leverage properties of the càdlàg function classes (Gill et al., 1995) and non-parametric higher-order spline Highly Adaptive Lasso (van der Laan, 2023) to establish equicontinuity conditions.

The rest of the paper is organized as follows. Section 2 presents the statistical setup and the general TMLE-OSLAD framework. Section 3 specializes the framework for CARA designs involving multiple surrogates. Section 4 provides theoretical results. Section 5 presents simulation studies, followed by a real-data-based simulation in Section 6. Section 7 concludes.

2 TMLE-Based Online Super Learner Adaptive Design

This section outlines the data structure, notation, and methods for the proposed TMLE-OSLAD framework. Suppose we have baseline covariates $W \in \mathcal{W}$ and a treatment $A \in \mathcal{A} = \{0, 1\}$. Let Y_K denote the primary outcome, observed K follow-up periods after treatment assignment, with larger values indicating better outcomes. We also consider $K - 1$ intermediate surrogate outcomes Y_k , each observed at $k = 1, \dots, K - 1$. We assume all outcomes Y_k lie in $[0, 1]$. Let T denote the total duration of enrollment in the experiment. At each time point, the experimenter uses historical data from previously enrolled participants to update the treatment assignment mechanism for new participants. These updates can aim to increase the chance that participants receive optimal personalized treatment by learning heterogeneous treatment effects, which are detailed in Section 3 and Appendix D.

2.1 Statistical Setup

Sequential data structure.

We use $t = 1, \dots, T$ to index the sequential enrollment times when participants enter the study and are randomized to one of the treatment arms. Let $E(t)$ denote the number of subjects enrolled at time t , and define the cumulative enrollment by $N(t) = \sum_{t'=1}^t E(t')$. For each participant i , we

sequentially observe $O_i = (t_i, W_i, A_i, Y_{i,1}, \dots, Y_{i,K})$, where t_i is the enrollment time, W_i are baseline covariates observed at t_i , and A_i is the treatment assigned at t_i . Surrogate outcomes $Y_{i,1}, \dots, Y_{i,K-1}$ are observed at times $t_i + k$ for $k = 1, \dots, K - 1$. The final outcome of interest, $Y_{i,K}$, is observed at $t_i + K$. To emphasize the timeline when each variable becomes available, define

$\tilde{W}_i(t_i) \equiv W_i, \tilde{A}_i(t_i) \equiv A_i$, and $\tilde{Y}_i(t_i + k) \equiv Y_{i,k}$ for $k = 1, \dots, K$. We use these notations interchangeably. At each time t , the experimenter has access to all information accrued up to $t - 1$, denoted by $\bar{O}(t - 1) := \{\tilde{W}_i(t'), \tilde{A}_i(t'), \tilde{Y}_i(t') : i \in [N(t - 1)], t' \leq t - 1\}$. This represents all covariates, treatments, and outcomes of previous participants that have become observable before time t . From $t - 1$ to t , new data are collected from two sets: $\mathcal{I}_1(t) := \{i : t - K \leq t_i \leq t - 1\}$ and $\mathcal{I}_2(t) := \{i : t_i = t\}$. $\mathcal{I}_1(t)$ contains previously enrolled participants whose outcomes are still arriving, and $\mathcal{I}_2(t)$ contains participants newly enrolled at time t . The experimenter first observes the outcomes revealed at time t for $\mathcal{I}_1(t)$, then updates the history

$\bar{\mathcal{H}}(t) := \bar{O}(t - 1) \cup \{\tilde{Y}_i(t) : i \in \mathcal{I}_1(t)\}$, where $\tilde{Y}_i(t)$ is the outcome for participant i that becomes observable at time t . Next, the experimenter enrolls $\mathcal{I}_2(t)$, observes their baseline covariates, and assigns treatments.

Adaptive treatment randomization. When a new participant i is enrolled at time t_i , a treatment A_i can be randomized based on the information available up to that point and the baseline covariates W_i . We denote the participant-specific pre-treatment history as $\mathcal{H}(i) := (\overline{\mathcal{H}}(t_i), W_i)$, where $\overline{\mathcal{H}}(t_i)$ is the observed data of previous participants enrolled before time t_i . We assume that treatment randomization depends on a fixed-dimensional summary context of the pre-treatment history, defined as $C_i := f_C(\mathcal{H}(i)) \in \mathcal{C}$. The treatment randomization probability is generated by a mapping $g_i : \mathcal{C} \rightarrow \mathcal{G}$, where $\mathcal{G}$ is a class of probability distributions of treatment assignment, and $g_i(a | C_i)$ is the probability of assigning $a \in \mathcal{A}$ given context C_i .

Likelihood.

Let Q_W denote the distribution of baseline covariates W and Q_{Y_k} denote the distribution of Y_k given $W, A, Y_1, \dots, Y_{k-1}$ for outcome Y_k ($k = 1, \dots, K$). We assume for every participant i , $t_i \perp\!\!\!\perp (W_i, A_i, Y_{i,1}, \dots, Y_{i,K})$, $W_i \sim Q_W$, and $Y_{i,k} \sim Q_{Y_k}(\cdot | W_i, A_i, Y_{i,1}, \dots, Y_{i,k-1})$. Specifically, let q_W denote the marginal density of the baseline covariates W , and q_{Y_k} denote the conditional density of the k -th outcome Y_k given $W, A, Y_1, \dots, Y_{k-1}$. The likelihood of the observed data of n subjects is given by $p^n(o_1, \dots, o_n) = \prod_{i=1}^n q_W(w_i) g_i(a_i | C_i) \prod_{k=1}^K q_{Y_k}(y_{i,k} | w_i, a_i, y_{i,1}, \dots, y_{i,k-1})$.

Structural Causal Model. We formalize the data generating process through a Structural Causal Model (SCM) (Pearl, 2000) over the experiment, which also underpins estimation and inference in our setting. At each time t , we first observe the history data $\overline{\mathcal{H}}(t)$, which includes, for previously enrolled participants with $t_i = t - k$ ($k = 1, \dots, K$), the outcomes given by $Y_{i,k} = f_{Y_k}(W_i, A_i, Y_{i,1}, \dots, Y_{i,k-1}, U_{Y_{i,k}})$. Second, for each new unit i enrolled at time t , we observe $t_i = t$; $W_i = f_W(U_{W_i})$; $C_i = f_C(\mathcal{H}(i))$; and $A_i = f_A(C_i, U_{A_i})$. Here, $U_i = (U_{W_i}, U_{A_i}, U_{Y_{i,1}}, \dots, U_{Y_{i,K}})$ are exogenous variables independent across participants. The actual adaptive design for n subjects is represented by the sequence of applied treatment-randomization functions, denoted by $\mathbf{g}_{1:n} := \{g_i : i = 1, \dots, n\}$.

2.2 Evaluation of Counterfactual CARA Designs

Here we define a target estimand within a modified SCM under a counterfactual adaptive design. Specifically, we replace every $A_i = f_A(C_i, U_{A_i})$ in the original SCM with $A_i^* \sim g_{*,i}(\cdot | C_i)$ for every i , which is a treatment randomization function under a counterfactual adaptive design, based on the observed summary measure C_i for each unit i at their entry time. The outcomes

then follow $Y_{i,k}^* = f_{Y_k}(W_i, A_i^*, Y_{i,1}^*, \dots, Y_{i,k-1}^*, U_{Y_{i,k}})$ for $k = 1, \dots, K$. Let $\bar{C}(n) := (C_1, \dots, C_n)$. Under this modified SCM, the proposed estimand is:

$$\Psi_{\mathbf{g}_{*,1:n}(\bar{C}(n))} = \frac{1}{n} \sum_{i=1}^n E_{g_{*,i}(\cdot|C_i)}[Y_{i,K}^*],$$

the mean counterfactual primary outcome had all participants, at their entry, been treated according to the treatment randomization mechanism $g_{*,i}$ that could have been applied under an alternative adaptive design $\mathbf{g}_{*,1:n} = (g_{*,1}, \dots, g_{*,n})$.

This target estimand naturally supports the counterfactual evaluation of multiple candidate adaptive designs of interest. Suppose we have J different candidate adaptive designs, where each design is defined by a series of counterfactual treatment randomization functions $\{g_{j,i}^* : i = 1, \dots, n\}$ for $j \in [J]$, and depends on the j -specific design configuration aiming to increase the probability that new participants receive personalized optimal treatment assignments to maximize their outcomes. The corresponding counterfactual estimand for the j -th candidate design is defined as: $\Psi_j := \frac{1}{n} \sum_{i=1}^n E_{g_{j,i}^*(\cdot|C_i)}[Y_{i,K}^*]$, reflecting the expected final outcome had the population been treated under that adaptive design. This enables a principled comparison of all candidate designs by comparing the values of $\{\Psi_j\}_{j=1}^J$.

Let $Q_0 := (Q_{0,W}, Q_{0,K}) \in \mathcal{Q}_W \times \mathcal{Q}_K$, where $Q_{0,W}$ is the true distribution of W , and $Q_{0,K}$ is the conditional distribution of Y_K given A and W , with the associated $\bar{Q}_{0,K}$ denoting the true conditional mean function of the final outcome Y_K given A and W . We also use $g_{0,i}$ to denote the actual treatment randomization function applied to each participant i in the experiment, which is known to the experimenter. One can identify the proposed causal estimand based on the observed data under the following assumptions.

Assumption 1 (Sequential Exchangeability).

For any $i \in [n]$, $Y_{i,K}(a) \perp\!\!\!\perp A_i | C_i$.

Assumption 2 (Positivity).

For any $i \in [n]$ and $a \in \mathcal{A}$, $g_{0,i}(a | C_i) > 0$.

Theorem 1 (Identification).

With assumptions 1 and 2, one can identify Ψ_j under Q_0 by

$$\Psi_j(Q_0) = \Psi_{\mathbf{g}_{j,ln}^*}(Q_0) = \frac{1}{n} \sum_{i=1}^n E_{Q_0, g_{j,i}^*} [Y_{i,K}^*] = \frac{1}{n} \sum_{i=1}^n \sum_{a=0}^1 \bar{Q}_{0,K}(a, W_i) g_{j,i}^*(a | C_i).$$

We note that this estimand provides a fast and robust approximation to an ideal estimand: the expected outcome obtained by integrating the entire distribution of data that would sequentially arise from the candidate adaptive design of interest. However, the latter's estimation is considerably more complex and computationally intensive. In contrast, our proposed context-specific estimand substantially simplifies the estimation procedure, enables doubly-robust estimation, and closely approximates the ideal estimand when the observed contexts are reflective of those that would be observed in the candidate design, especially when the adaptation procedure is based on learning from a stationary target (e.g. Q_0) that dominates over the experiments. This yields an efficient and effective approach to distinguishing high-performing from low-performing designs and facilitating online selection among candidate designs. We refer readers to Appendix A.2 for further discussions and empirical results showing minimal difference between the two estimands in our simulations.

2.3 Targeted Maximum Likelihood Estimation

At each time point t , let $n_t := N(t - K)$ denote the total number of participants for whom the final outcome Y_K has been observed at that time. We define the target estimand for the j -th counterfactual adaptive design at time t as

$$\Psi_{t,j} := \frac{1}{n_t} \sum_{i=1}^{n_t} E_{Q_0, g_{j,i}^*} [Y_{i,K}^*] = \frac{1}{n_t} \sum_{i=1}^{n_t} \sum_{a=0}^1 \bar{Q}_{0,K}(a, W_i) g_{j,i}^*(a | C_i). \text{ To estimate } \Psi_{t,j}, \text{ we provide a}$$

Targeted Maximum Likelihood Estimation (TMLE) approach. Let $\bar{Q}_{n,K,t}$ be an initial estimate of $\bar{Q}_{0,K}$ based on the historical data $\bar{\mathcal{H}}(t)$ observed up to time t . We apply a TMLE procedure to update $\bar{Q}_{n,K,t}$ by fitting the logistic model (van der Laan and Rubin, 2006; van der Laan, 2008; van der Laan and Rose, 2011; Malenica et al., 2024): $\text{logit} \bar{Q}_{n,K,t,\epsilon}(A, W) = \text{logit} \bar{Q}_{n,K,t}(A, W) + \epsilon$, with weights $\frac{g_{j,i}^*(A_i | C_i)}{g_{0,i}(A_i | C_i)}$. The estimate of ϵ is denoted by $\hat{\epsilon}_t$, and an updated estimator of the

conditional mean function $\bar{Q}_{n,K,t}^* := \bar{Q}_{n,K,t,\hat{\epsilon}_t}$ is obtained, which solves:

$$\frac{1}{n_t} \sum_{i=1}^{n_t} \frac{g_{j,i}^*(A_i | C_i)}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_{n,K,t}^*(A_i, W_i)) = 0. \text{ The TMLE estimate of } \Psi_{t,j} \text{ is:}$$

$$\hat{\Psi}_{t,j} := \frac{1}{n_t} \sum_{i=1}^{n_t} \sum_{a=0}^1 \bar{Q}_{n,K,t}^*(a, W_i) g_{j,i}^*(a | C_i).$$

The estimated standard error is $\sqrt{\hat{\sigma}_{t,j}^2 / n_t}$, where

$$\hat{\sigma}_{t,j}^2 := \frac{1}{n_t} \sum_{i=1}^{n_t} \left[\frac{g_{j,i}^*(A_i | C_i)}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_{n,K,t}^*(A_i, W_i)) \right]^2. \text{ The } 100(1-\alpha)\% \text{ Wald confidence interval for}$$

$\hat{\Psi}_{t,j}$ is $\left[\hat{\Psi}_{t,j} - z_{1-\alpha/2} \sqrt{\frac{1}{n_t} \hat{\sigma}_{t,j}^2}, \hat{\Psi}_{t,j} + z_{1-\alpha/2} \sqrt{\frac{1}{n_t} \hat{\sigma}_{t,j}^2} \right]$, where $z_{1-\alpha/2}$ is the $1-\alpha/2$ quantile of the standard normal distribution (e.g. $\alpha = 0.05$).

Under the conditions in Section 4, TMLE is consistent, asymptotically normal and doubly robust, meaning that consistency holds if the outcome regression or the treatment assignment mechanism is correctly specified. In adaptive designs, the treatment assignment mechanism is known, so TMLE remains consistent even if the outcome regression is misspecified. That said, more accurate estimation of $\bar{Q}_{0,K}$ can reduce the variance of TMLE. When multiple estimators are available, a Super Learner (van der Laan et al., 2007) can be used to select the best-performing estimator by cross-validation. We further recommend including the higher-order Highly Adaptive Lasso (van der Laan, 2023) to facilitate the equicontinuity conditions.

2.4 Online Super Learner Adaptive Design

As the experiment progresses to a new time point t , there are up to J candidate treatment randomization functions $g_{1,i}^*, \dots, g_{J,i}^*$ for assigning treatment randomization probabilities for a new participant i enrolled at t . For each $j \in [J]$, $g_{j,i}^*$ is generated based on historical data using the j -specific adaptive design configurations. The proposed Online Super Learner Adaptive Design begins by estimating $\hat{\Psi}_{t,j}$ for each $j \in [J]$ and obtaining confidence intervals for these

estimates. Then, it selects the design indexed by $j_t^* := \operatorname{argmax}_{j \in [J]} \hat{\Psi}_{t,j} - z_{1-\alpha/2} \sqrt{\frac{1}{n_t} \hat{\sigma}_{t,j}^2}$, the candidate

design with the highest lower confidence interval bound among all $\hat{\Psi}_{t,j}$ for $j \in [J]$, prioritizing designs with high performance with low uncertainty to mitigate the risk of selecting a suboptimal design due to unstable estimates. Finally, for all participants i enrolled at time t , we assign treatment randomization probabilities using $g_{j_t^*,i}^*$.

2.5 Statistical Inference at the End of Adaptive Experiments

We denote by $T_{\text{end}} := T + K$ the time when the experiment concludes, i.e., when all final outcomes Y_K have been observed after the last enrollment at time T . The corresponding sample

size is $n_{\text{end}} := N(T) = \sum_{t=1}^T E(t)$. At time T_{end} , the TMLE procedure in Section 2.3 applies to

estimate $\Psi_{T_{\text{end}},j}$, the expected final outcome under the j th counterfactual adaptive design. We also provide TMLEs for other causal estimands commonly studied in randomized trials,

including the average treatment effect and the mean outcome under dynamic treatment rules. Their statistical inference is based on Wald confidence intervals using the corresponding TMLE estimates and their estimated standard errors.

Average treatment effect. We define the following average treatment effect (ATE) on the primary outcome Y_K across all participants within the experiment:

$$\begin{aligned} \psi_{T_{end}}^{ATE} &:= \frac{1}{n_{end}} \sum_{i=1}^{n_{end}} [\bar{Q}_{0,K}(1, W_i) - \bar{Q}_{0,K}(0, W_i)]. \text{ For estimation, we first obtain an initial estimate } \bar{Q}_{n,K,T_{end}} \text{ of } \bar{Q}_{0,K} \text{ and then perform the TMLE update with weights } \frac{2A_i - 1}{g_{0,i}(A_i | C_i)}, \text{ yielding } \bar{Q}_{n,K,T_{end}}^* \\ \text{that solves } &\frac{1}{n_{end}} \sum_{i=1}^{n_{end}} \frac{2A_i - 1}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_{n,K,T_{end}}^*(A_i, W_i)) = 0. \text{ The TMLE estimate of } \psi_{T_{end}}^{ATE} \text{ is} \\ \hat{\psi}_{T_{end}}^{ATE} &:= \frac{1}{n_{end}} \sum_{i=1}^{n_{end}} [\bar{Q}_{n,K,T_{end}}^*(1, W_i) - \bar{Q}_{n,K,T_{end}}^*(0, W_i)]. \text{ The standard error estimate is } \sqrt{\hat{\sigma}_{T_{end}}^{2,ATE} / n_{end}}, \\ \text{where } \hat{\sigma}_{T_{end}}^{2,ATE} &= \frac{1}{n_{end}} \sum_{i=1}^{n_{end}} \left[\frac{2A_i - 1}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_{n,K,T_{end}}^*(A_i, W_i)) \right]^2. \end{aligned}$$

Mean outcome under a dynamic treatment rule. Let $d: \mathcal{W} \rightarrow \mathcal{A}$ denote a fixed dynamic treatment rule mapping baseline covariates to a treatment arm. We define a target estimand

$$\begin{aligned} \psi_{T_{end}}^d &:= \frac{1}{n_{end}} \sum_{i=1}^{n_{end}} \bar{Q}_{0,K}(d(W_i), W_i) \text{ as the mean final outcome } Y_K \text{ under rule } d. \text{ We apply TMLE with} \\ \text{weights } &\frac{I(A_i = d(W_i))}{g_{0,i}(A_i | C_i)} \text{ to solve } \frac{1}{n_{end}} \sum_{i=1}^{n_{end}} \frac{I(A_i = d(W_i))}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_{n,K,T_{end}}^*(A_i, W_i)) = 0, \text{ and yield} \\ \hat{\psi}_{T_{end}}^d &:= \frac{1}{n_{end}} \sum_{i=1}^{n_{end}} \bar{Q}_{n,K,T_{end}}^*(d(W_i), W_i). \text{ The standard error estimate is } \sqrt{\hat{\sigma}_{T_{end}}^{2,d} / n_{end}}, \text{ where} \\ \hat{\sigma}_{T_{end}}^{2,d} &:= \frac{1}{n_{end}} \sum_{i=1}^{n_{end}} \left[\frac{I(A_i = d(W_i))}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_{n,K,T_{end}}^*(A_i, W_i)) \right]^2. \end{aligned}$$

We also note their marginal versions $\psi^{ATE} := E[\bar{Q}_{0,K}(1, W) - \bar{Q}_{0,K}(0, W)]$ and $\psi^d := E[\bar{Q}_{0,K}(d(W), W)]$. The TMLE estimates are the same as above, while the standard error estimates differ, with $\sqrt{\hat{\sigma}^{2,ATE} / n_{end}}$ for ψ^{ATE} and $\sqrt{\hat{\sigma}^{2,d} / n_{end}}$ for ψ^d , where (Tex translation failed) and

$$\hat{\sigma}^{2,d} := \frac{1}{n_{end}} \sum_{i=1}^{n_{end}} \left[\frac{I(A_i = d(W_i))}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_{n,K,T_{end}}^*(A_i, W_i)) + \bar{Q}_{n,K,T_{end}}^*(d(W_i), W_i) - \hat{\psi}^d \right]^2.$$

3 Evaluating and Utilizing Surrogates in CARA Designs

In this section, we apply the TMLE-OSLAD framework to the motivating example of adaptive experiments involving multiple surrogate outcomes before the final outcome. Recall that there are $K - 1$ candidate surrogate outcomes, $Y_1, \dots, Y_{K-1}$ measured before the primary outcome Y_K at follow-up times $k = 1, \dots, K - 1$. Each surrogate Y_k defines a distinct candidate adaptive design, which leverages accumulating information from short-term surrogate outcomes to increase the probability for assigning the estimated personalized treatment associated with that surrogate over time.

TMLE-OSLAD is implemented to continuously evaluate and dynamically select among different surrogate-specific adaptive designs over the experiment. We first introduce a CARA procedure that adapts treatment randomization probabilities based on the estimated conditional average treatment effect (CATE) to improve participant outcomes. Building on Section 2, we define our target estimand that evaluates each candidate surrogate within a counterfactual adaptive design that utilizes the surrogate to adapt treatment randomization probabilities. We then demonstrate TMLE-OSLAD that uses TMLE to estimate these target estimands to evaluate surrogate-guided adaptive designs, and utilizes the most useful one to guide the ongoing adaptive experiment. Finally, we conduct post-experiment inference for other causal estimands such as ATE and the mean primary outcome under optimal dynamic treatment rules that optimize primary or surrogate outcomes.

3.1 CARA Designs Using Surrogate Outcomes

In our CARA designs, at each time point, we update the model to estimate the CATE for the outcome of interest and its associated uncertainty, and use these estimates to assign randomization probabilities to newly enrolled participants. This design aims to maximize participant outcomes by increasing the chance to allocate the estimated optimal treatment, with a principled exploration–exploitation balance ensuring sufficient exploration when the superiority of a treatment arm is uncertain and maintaining the ability to continue learning heterogeneous treatment effect from data collected at later time points. Specifically, when the estimated CATE is small relative to its uncertainty, indicating that the superiority of one treatment arm is uncertain, the treatment randomization probabilities remain close to 0.5. This avoids overcommitting to a noisy and potentially suboptimal rule and ensures sufficient exploration across treatment arms. As the estimated CATE suggests a stronger signal for the superior treatment arm, the probability of assigning that treatment increases up to a maximum exploitation probability. Meanwhile, a positive lower bound on the probability of assigning the alternative arm is maintained to ensure sufficient support for valid estimation and inference of other estimands at the end of the experiment (see Sections 2.5 and 3.3).

More details of the CARA design are provided in the Appendix. Specifically, CATE is estimated using a doubly robust approach based on pseudo-outcomes (van der Laan, 2006; Luedtke and van der Laan, 2016b) (Appendix C), with point estimates and confidence intervals constructed based on higher-order Highly Adaptive Lasso (HAL) (van der Laan, 2023) (Appendix B). Details of the treatment randomization function can be found in Appendix D.

3.2 Surrogate-Guided Online Super Learner Adaptive Design

To enable data-adaptive selection of the most effective surrogate to guide treatment allocation, we leverage the general TMLE-OSLAD framework in Section 2 to both evaluate and dynamically utilize candidate surrogate outcomes within real-time CARA experiments. At each interim time t , we use all observed historical data to assess the utility of each surrogate-driven adaptive design and to select the most promising surrogate for adaptation.

First, at time t , we define the target estimand $\Psi_{t, \mathbf{g}_{k, \cdot}^*} := \frac{1}{n_t} \sum_{i=1}^{n_t} E_{g_{k, \cdot}^*} [Y_{i, K}^*]$ for each candidate surrogate Y_k , which represents the mean final outcome that would be achieved for the n_t fully-followed participants if each had been assigned with the treatment randomization function $g_{k, \cdot}^*$, at their enrollment, determined by the adaptive design using surrogate Y_k . We refer to this estimand as “the utility of a surrogate” in the CARA design (see Appendix A.1 for further discussion of this estimand).

To construct $g_{k, \cdot}^*$ from the CATE for Y_k , we impose the following identification assumption.

Assumption 3 (Sequential Exchangeability for Surrogate Outcomes).

For any $k \in \{1, \dots, K-1\}$ and any subject $i \in \{1, \dots, n_t\}$, $Y_{i, k}(a) \perp\!\!\!\perp A_i \mid C_i$.

Under Assumptions 1 and 2, $\Psi_{t, \mathbf{g}_{k, \cdot}^*}$ can be identified as $\Psi_{t, k} := \frac{1}{n_t} \sum_{i=1}^{n_t} \sum_{a=0}^1 \bar{Q}_{0, K}(a, W_i) g_{k, i}^*(a \mid C_i)$.

Then, we estimate $\Psi_{t, k}$ for each surrogate Y_k using TMLE, which is given by

$\hat{\Psi}_{t, k} = \frac{1}{n_t} \sum_{i=1}^{n_t} \sum_{a=0}^1 \bar{Q}_{n, K, t}^*(a, W_i) g_{k, i}^*(a \mid C_i)$. The estimated standard error is $\sqrt{\hat{\sigma}_{t, k}^2 / n_t}$, where

$\hat{\sigma}_{t, k}^2 = \frac{1}{n_t} \sum_{i=1}^{n_t} \left[\frac{g_{k, i}^*(A_i \mid C_i)}{g_{0, i}(A_i \mid C_i)} (Y_{i, K} - \bar{Q}_{n, K, t}^*(A_i, W_i)) \right]^2$. Next, we compare all candidate surrogates and

the final outcome by constructing confidence intervals for $\Psi_{t, k}$ ($k = 1, \dots, K$) and identify the

outcome $Y_{k_t}^*$ with the highest lower confidence bound, where $k_t^* := \arg\max_{k \in [K]} \hat{\Psi}_{t, k} - z_{1-\alpha/2} \sqrt{\hat{\sigma}_{t, k}^2 / n_t}$.

For example, if two surrogates have similar point estimates but one yields a much narrower confidence interval, the surrogate with the higher lower bound is prioritized, as it reflects more evidence in its utility relative to uncertainty. Accordingly, we select $Y_{k_t}^*$ as the outcome to guide the next-stage treatment assignment for newly enrolled participants at time t , utilizing the treatment randomization function $g_{k_t, \cdot}^*$. This procedure is repeated at each subsequent time point, allowing adaptive surrogate selection and treatment allocation as new data accrue.

3.3 Inference after Adaptive Experiments Involving Surrogates

At the end of the adaptive experiment, one can follow Section 2.5 to conduct statistical inference for causal estimands such as the average treatment effect or the mean outcome under dynamic treatment rules. Here, we consider the mean primary outcome under optimal dynamic treatment rules (ODTR) that maximize the primary outcome (Murphy, 2003; Robins, 2004; van der Laan and Luedtke, 2015) or a surrogate outcome (Hsu et al., 2015; Yang et al., 2024).

For a primary or surrogate outcome Y_k ($k \in \{1, \dots, K\}$), define the optimal dynamic treatment rule $d_{0,k}^* : \mathcal{W} \rightarrow \mathcal{A}$ by $d_{0,k}^*(W) = I(E[Y_k(1) - Y_k(0) | W] > 0)$. We estimate this rule using all observed data and obtain $d_{n,k,T_{\text{end}}}^*$ (for brevity, we also write $d_{n,k}^*$). The target estimand is the mean primary outcome Y_K under this rule $\psi_{T_{\text{end}}}^{d_{n,k}^*} := \frac{1}{n_{\text{end}}} \sum_{i=1}^{n_{\text{end}}} \bar{Q}_{0,K}(d_{n,k}^*(W_i), W_i)$, or its marginal version $\psi^{d_{n,k}^*} := E[\bar{Q}_{0,K}(d_{n,k}^*(W), W)]$. We estimate them by TMLE solving

$$\frac{1}{n_{\text{end}}} \sum_{i=1}^{n_{\text{end}}} \frac{I(A_i = d_{n,k}^*(W_i))}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_{n,K,T_{\text{end}}}^*(A_i, W_i)) = 0, \text{ yielding a point estimate}$$

$$\hat{\psi}_{T_{\text{end}}}^{d_{n,k}^*} := \frac{1}{n_{\text{end}}} \sum_{i=1}^{n_{\text{end}}} \bar{Q}_{n,K,T_{\text{end}}}^*(d_{n,k}^*(W_i), W_i), \text{ with standard error estimates given by Section 2.5.}$$

We note that this approach provides an additional treatment-rule-based perspective on evaluating surrogate utility by the mean primary outcome Y_K achieved under ODTRs maximizing a surrogate outcome Y_k ($k \neq K$), which can be compared with the value achieved under the ODTR maximizing Y_K itself. Moreover, for $d_{n,K}^*$, the estimated ODTR maximizing the primary outcome Y_K , the standard TMLE for its evaluation may exhibit upward bias as the same data are used to both learn the rule and evaluate its value. We recommend applying cross-validated TMLE (CV-TMLE) (Zheng and van der Laan, 2011; van der Laan and Luedtke, 2015; Montoya et al., 2021), which uses sample splitting to avoid finite sample bias.

4 Theoretical Analysis of TMLE

This section derives consistency and asymptotic normality of the proposed TMLE. For notational continuity with the surrogate-guided adaptive design setup introduced above, we continue to index candidate designs by $k \in [K]$ in the TMLE analysis, and the target estimand of interest is $\Psi_{t,k}$. This indexing is notationally equivalent to the general index $j \in [J]$ used in Section 2 for estimating $\Psi_{t,j}$; the theoretical results and proofs apply identically under either notation. We first study the difference between TMLE $\hat{\Psi}_{t,k}$ and the truth $\Psi_{t,k}$. For any $\bar{Q}_K \in \bar{\mathcal{Q}}_K$, define

$$D_k(\bar{Q}_K)(O_i, C_i) := \frac{g_{k,i}^*(A_i | C_i)}{g_{0,i}(A_i | C_i)} (Y_{i,K} - \bar{Q}_K(A_i, W_i)).$$

Theorem 2 .

For any $\bar{Q}_{n,K,t}^* \in \bar{\mathcal{Q}}_K$ and $\bar{Q}_{1,K} \in \bar{\mathcal{Q}}_K$, the difference between $\hat{\Psi}_{t,k}$ and $\Psi_{t,k}$ can be decomposed as $\hat{\Psi}_{t,k} - \Psi_{t,k} = M_{1,n_t}(\bar{Q}_{1,K}) + M_{2,n_t}(\bar{Q}_{n,K,t}^*, \bar{Q}_{1,K})$, where

$$\begin{aligned} M_{1,n_t}(\bar{Q}_{1,K}) &= \frac{1}{n_t} \sum_{i=1}^{n_t} \left[D_k(\bar{Q}_{1,K})(O_i, C_i) - E \left[D_k(\bar{Q}_{1,K})(O_i, C_i) \mid \mathcal{H}(i) \right] \right], \\ M_{2,n_t}(\bar{Q}_{n,K,t}^*, \bar{Q}_{1,K}) &= \frac{1}{n_t} \sum_{i=1}^{n_t} \left\{ \left[D_k(\bar{Q}_{n,K,t}^*)(O_i, C_i) - E \left[D_k(\bar{Q}_{n,K,t}^*)(O_i, C_i) \mid \mathcal{H}(i) \right] \right] \right. \\ &\quad \left. - \left[D_k(\bar{Q}_{1,K})(O_i, C_i) - E \left[D_k(\bar{Q}_{1,K})(O_i, C_i) \mid \mathcal{H}(i) \right] \right] \right\}. \end{aligned}$$

The difference of TMLE $\hat{\Psi}_{t,k}$ and the true $\Psi_{t,k}$ is decomposed into M_{1,n_t} , an average of a martingale difference sequence, and a martingale process M_{2,n_t} . Under the strong positivity and stabilized conditional variance conditions below, the asymptotic normality of M_{1,n_t} follows directly from the martingale central limit theorem (e.g., Theorem 2 in Brown (1971)).

Assumption 4 (Strong positivity).

For any i , there exists $\zeta > 0$ such that $g_{0,i} > \zeta$ and $g_{k,i}^* > \zeta$.

Assumption 5 (Stabilization of the average of conditional variances).

There exists $\sigma_{t,k}^2 \in \mathbb{R}^+$ such that $\frac{1}{n_t} \sum_{i=1}^{n_t} \text{Var} \left[D_k(\bar{Q}_{1,K})(O_i, C_i) \mid \mathcal{H}(i) \right] \xrightarrow{p} \sigma_{t,k}^2$.

Theorem 3 (Asymptotic normality of the first term).

Suppose that assumptions 4 and 5 hold, then $\sqrt{n_t} M_{1,n_t}(\bar{Q}_{1,K}) \xrightarrow{d} N(0, \sigma_{t,k}^2)$.

We note that M_{1,n_t} depends only on a fixed function $\bar{Q}_{1,K} \in \bar{\mathcal{Q}}_K$. For asymptotic normality of the TMLE, $\bar{Q}_{1,K}$ need not equal the true conditional mean $\bar{Q}_{0,K}$ as long as $\bar{Q}_{n,K,t}^*$ converges to $\bar{Q}_{1,K}$ satisfying the equicontinuity condition for the second term, M_{2,n_t} , as formalized below.

Assumption 6 (Entropy integral of $\bar{\mathcal{Q}}_K$).

Let $N_\infty(\epsilon, \bar{\mathcal{Q}}_K)$ denote the ϵ -covering number in supremum norm of $\bar{\mathcal{Q}}_K$. Define

$$J_\infty(\epsilon, \bar{\mathcal{Q}}_K) := \int_0^\epsilon \sqrt{\log(1 + N_\infty(u, \bar{\mathcal{Q}}_K))} du \text{ as the } \epsilon\text{-entropy integral in supremum norm of } \bar{\mathcal{Q}}_K.$$

Assume that $\mathcal{J}_\infty(\epsilon, \bar{\mathcal{Q}}_K)$ is finite and $\mathcal{J}_\infty(\epsilon, \bar{\mathcal{Q}}_K) \rightarrow 0$ as $\epsilon \rightarrow 0$.

Assumption 7 (Convergence of $\bar{Q}_{n,K,t}^*$).

For any $\bar{Q}_K \in \bar{\mathcal{Q}}_K$, define

$$s_{n_t}(\bar{Q}_K, \bar{Q}_{1,K}) := \sqrt{\frac{1}{n_t} \sum_{i=1}^{n_t} E \left[\phi \left(\frac{|D_k(\bar{Q}_K)(O_i, C_i) - D_k(\bar{Q}_{1,K})(O_i, C_i)|}{\nu} \right) \middle| C_i \right]},$$

where $\phi(x) := e^x - x - 1$, $\nu := 8(1 - \zeta) / \zeta$. Assume that $s_{n_t}(\bar{Q}_{n,K,t}^*, \bar{Q}_{1,K}) = O_p(\delta_{n_t})$ for some $\delta_{n_t} \rightarrow 0$.

Theorem 4 (Equicontinuity).

Under Assumptions 4, 6, and 7, $M_{2,n_t}(\bar{Q}_{n,K,t}^*, \bar{Q}_{1,K}) = o_p\left(n_t^{-\frac{1}{2}}\right)$.

This equicontinuity result demonstrates that, in addition to the asymptotic normality of M_{1,n_t} , the remainder term M_{2,n_t} is negligible. The proof is given in Appendix F, which depends on the equicontinuity result of the martingale process in general setup with sequential dependence (Appendix E). We further provide a sufficient way to satisfy the equicontinuity condition by leveraging the recent work of van der Laan (2023) that introduces a class of m -th order smoothness class of functions with m -th-order Radon-Nikodym derivative w.r.t. Lebesgue measure is càdlàg and of bounded variation ($m = 1, 2, \dots$), along with a higher-order spline Highly Adaptive Lasso (HAL) estimator (see Appendix B). We show that if $\bar{Q}_{n,K,t}$ is estimated by higher-order HAL and converges to a $\bar{Q}_{1,K}$ that is in the same class of functions, then the equicontinuity condition holds.

Assumption 8.

$\bar{Q}_{1,K}$ and $\bar{Q}_{n,K,t}$ are both from a class of functions that are represented by an m -th order smooth function whose m -th order Radon-Nikodym derivatives are càdlàg functions with bounded sectional variation norm ($m = 1, 2, \dots$), and $\bar{Q}_{n,K,t}$ is obtained by the m -th order Highly Adaptive Lasso estimator.

Corollary 1.

Suppose Assumptions 4 and 8 hold. Then $M_{2,n_t}(\bar{Q}_{n,K,t}^*, \bar{Q}_{1,K}) = o_p\left(n_t^{-\frac{1}{2}}\right)$.

Theorem 5 (Asymptotic normality of TMLE).

Suppose Assumptions 4, 5, 6, and 7 hold, or alternatively, Assumptions 4, 5, and 8 hold, then

$$\sqrt{n_t}(\hat{\Psi}_{t,k} - \Psi_{t,k}) \xrightarrow{d} N(0, \sigma_{t,k}^2).$$

The asymptotic normality of TMLE is obtained directly from Theorems 3 and 4. We estimate the

standard error by $\hat{\sigma}_{t,k}^2 := \frac{1}{n_t} \sum_{i=1}^{n_t} \left\{ \frac{g_{k,i}^*(A_i | C_i)}{g_{0,i}(A_i | C_i)} [Y_{i,K} - \bar{Q}_{n,K,t}^*(A_i, W_i)] \right\}^2$ (which converges in

probability to $\sigma_{t,k}^2$), with the confidence interval $\left[\hat{\Psi}_{t,k} - z_{1-\alpha/2} \sqrt{\hat{\sigma}_{t,k}^2 / n_t}, \hat{\Psi}_{t,k} + z_{1-\alpha/2} \sqrt{\hat{\sigma}_{t,k}^2 / n_t} \right]$.

5 Simulation Studies

Simulation setup.

We present simulation studies to evaluate the performance of our proposed adaptive design and estimation methods. We consider a binary treatment $A \in \{0,1\}$, a univariate baseline covariate W following a uniform distribution $U(-4, 4)$, four surrogate outcomes Y_k measured at follow-up times $k = 1, 2, 3, 4$, respectively, and a final outcome Y_5 . Each participant's outcome Y_k is generated by its conditional mean $E(Y_k | A, W)$ plus a random error following the normal distribution ($\sim N(0,1)$). For detailed information on the simulations, we refer readers to Appendix C for CATE estimation, Appendix D for treatment randomization mechanism and Appendix G for data generating distributions. We evaluate the performance of various designs in two scenarios. In Scenario 1 (Figure 1a), the optimal dynamic treatment rule based on the final outcome is more aligned with that defined in terms of later versus earlier surrogates. This implies that, despite requiring some time to measure and thus delaying adaptation in randomization probabilities, later surrogates are more useful in guiding treatment assignment probabilities. In Scenario 2 (Figure 1b), all surrogate outcomes have the same ODTR as that of the final outcome. Notably, early surrogates have larger absolute CATE compared to later surrogates, suggesting that early surrogates are more sensitive to different treatments and potentially serve as superior indicators of the most beneficial treatment compared to the final outcome. In other words, in this scenario, earlier surrogates have the joint advantage of both timeliness and improved sensitivity for detecting treatment effect heterogeneity.

We compare seven distinct designs with varying treatment randomization mechanisms over 500 Monte Carlo runs: (a) a simple non-adaptive randomized controlled trial (RCT) design, which allocates two treatments with equal probability at each time point; (b) five adaptive designs with randomization probabilities updated in response to an *a priori* fixed choice of either a surrogate

outcome Y_k ($k = 1, \dots, 4$) or the final outcome Y_K ($K = 5$); (c) the proposed TMLE-OSLAD adaptive design that dynamically evaluates and selects which outcome (among $Y_1, \dots, Y_5$) is used to update treatment randomization probabilities at each time point. For each design, we run experiments for $T = 50$ time points with 50 subjects enrolled per time point. The maximal treatment assignment probability is capped at 90%.

Surrogate evaluation and TMLE performance. We first evaluate if TMLE-OSLAD can consistently identify the oracle surrogate, which we define as the candidate with the highest true $\Psi_{t,k}$ at time t among all Y_k (in other words, the surrogate that if used would result in the largest expected primary outcomes for all enrolled participants in the experiment). In particular, if $\Psi_{t,K}$ is the largest, then the primary outcome Y_K itself is the oracle. Table 1 presents the average of true $\Psi_{t,k}$ values at illustrative time points 11, 21, 31, 41, and 50 for both scenarios across Monte Carlo runs. The expected outcome of Y_5 under the non-adaptive randomization is also reported, denoted by $\Psi_{t,RCT}$. In Scenario 1, the oracle surrogate gradually shifts towards later surrogates as time elapses and more data from these surrogates become available to guide the adaptation of treatment randomization probabilities. Initially, surrogates Y_3 and Y_4 have the highest value at times 11 and 21, reflecting the balance they achieve between earlier availability and degree to which they reflect the true optimal individualized treatment assignment. While these surrogates provide benefit in the short term, they do not consistently predict the correct optimal treatment for the primary outcome for all individuals, leading them to be eventually outperformed by designs that utilize the true primary outcome Y_5 as additional follow-up time is accrued. This is expected, as the ultimate assessment of a surrogate's effectiveness as follow-up time extends hinges on its ability to predict the optimal personalized treatment for the primary outcome. In Scenario 2, as expected, surrogate Y_1 remains the most useful (i.e. is the oracle surrogate) throughout, due to its early availability, and the fact that it correctly identifies the true optimal individualized treatment for the long-term primary outcome. In contrast, later surrogates Y_4 and Y_5 have lower scores. This disparity can be attributed to the fact that while Y_4 and Y_5 will eventually serve equally well to identify the optimal personalized treatment as time since trial initiation goes to infinity, their utility is limited under finite trial durations because they have been observed for fewer participants. Across both scenarios, the TMLE estimates for these estimands exhibit low bias and variance, and their confidence intervals obtain nominal coverage (see Tables S1–S3 in Appendix G).

Figures 2a–2b show the frequency with which each candidate design is selected by TMLE-OSLAD over time in Scenarios 1 and 2, respectively, while Figures 2c–2d present a single-run example illustrating the selection mechanism. Without prior knowledge of relationships between surrogate and primary outcomes, the TMLE-OSLAD procedure is able to identify the oracle surrogate in each setting. In Scenario 1, where later surrogate outcomes serve as better proxies for the final outcome when determining optimal individualized treatments, TMLE-OSLAD progressively shifts its selections toward these later surrogates, ultimately favoring Y_4 and Y_5 most frequently. In contrast, in Scenario 2 where the earliest surrogate Y_1 remains the oracle

surrogate throughout, TMLE-OSLAD predominantly selects early surrogates, with Y_1 most frequently selected, followed by Y_2 and Y_3 , while Y_4 and Y_5 are rarely chosen.

Performance of benefiting participants. We use $d_{0,K} : \mathcal{W} \rightarrow \mathcal{A}$ to denote the true optimal individualized treatment rule for Y_K , which is defined as

$d_{0,K}(W) := I(\bar{Q}_{0,K}(1, W) - \bar{Q}_{0,K}(0, W) > 0)$. This rule assigns to each individual the treatment that would maximize their expected final outcome, given their covariates. To evaluate the extent to which each design assigns treatments that match the underlying oracle optimal rule, we define probability (%) of not assigning optimal individualized treatments as

$$p_t^{non-opt} := \frac{1}{E(t)} \sum_{i:t_i=t} I(A_i \neq d_{0,K}(W_i)), \text{ where } E(t) \text{ is the number of participants enrolled at time } t.$$

Similarly, $p_{t,RCT}^{non-opt}$ and $p_{t,SL}^{non-opt}$ denote this probability for the non-adaptive RCT and the proposed TMLE-OSLAD adaptive design, respectively. Table S4 in Appendix G compares these suboptimal treatment assignment probabilities across different designs. We also define the average regret for participants enrolled at time t within an adaptive design as

$$R_t := \frac{1}{E(t)} \sum_{i:t_i=t} I(A_i \neq d_{0,K}(W_i)) \times |\bar{Q}_{0,K}(1, W_i) - \bar{Q}_{0,K}(0, W_i)|. \text{ Here, each participant } i \text{ contributes}$$

the absolute difference in expected final outcomes between the optimal and actual treatments. A lower R_t indicates a better design that brings benefit in Y_K for participants enrolled at time t . We use $R_{t,k}$ to denote the average regret within an adaptive design responsive to Y_k at time t , while $R_{t,RCT}$ and $R_{t,SL}$ represent the average regret for the RCT design and TMLE-OSLAD design, respectively. Table S5 reports average regret across simulations at $t = 11, 21, 31, 41$, and 50 , with full trajectories shown in Figure S1.

In Scenario 1, where later surrogates are superior at identifying the optimal individualized treatment, the adaptive design responsive to Y_5 , as expected, consistently demonstrates the lowest regret across the experiment. In contrast, in Scenario 2, the regret within the adaptive design that responds to Y_1 remains consistently the lowest. There is a noticeable trend that the adaptive design in response to the final outcome Y_5 exhibits a slower convergence toward the regret of the oracle surrogate Y_1 , with a substantial discrepancy in regret that persists even at $t = 50$. Of course, in practice, the oracle surrogate is not known *a priori*, motivating the use of the TMLE-OSLAD procedure to evaluate and select among candidate surrogates in adaptive design. Across both scenarios, the proposed adaptive design demonstrates stable efficacy in approximating the minimal achievable regret compared to other designs. Its regret ranks second lowest starting at $t = 23$ in Scenario 1, and matches the lowest regret trajectory of Y_1 following $t = 21$ in Scenario 2. This robust performance reflects the advantage of the real-time evaluation and selection mechanism in TMLE-OSLAD, which continuously assesses each candidate surrogate-guided adaptive design, selects the most advantageous one to guide treatment randomization in subsequent stages.

Post-experiment inference for other causal estimands. Table 3 reports TMLE performance in estimating the additional causal estimands introduced in Section 3.3, including the average treatment effect and the mean final outcome under the estimated optimal dynamic treatment rules that maximize surrogate outcomes and the primary outcome. The mean final outcome Y_5 under the estimated optimal dynamic treatment rule for Y_5 is estimated using five-fold CV-TMLE.

Across all estimands, TMLE provides robust performance with low bias and variance, and nominal coverage of its confidence intervals.

6 Application

We conduct an extended simulation study inspired by the Adaptive Strategies for Preventing and Treating Lapses of Retention in HIV Care (ADAPT-R) trial (Geng et al., 2023), a sequential multiple assignment randomized (SMART) trial designed to evaluate adaptive strategies for improving retention in care among individuals initiating HIV treatment. We demonstrate our proposed TMLE-OSLAD framework in this context, showing how it utilizes effective surrogates to benefit participants and evaluating TMLE performance in this setup.

Simulation setup. We focus on Phase 2 of the ADAPT-R trial, where participants who had a lapse in care in Phase 1 (defined as missing a clinic visit by 14 or more days) were re-randomized to a re-engagement strategy: either a peer navigation strategy (Navigator) or a combined Short Message Service reminder of appointments and conditional cash transfer (SMS+CCT). We discretize the study timeline into 50-day stages and define outcomes as the proportion of time in care during which a participant is engaged in across five windows, from the first day of receiving the re-engagement strategy to five specific time points: 50, 100, 150, 200, and 250 days. These outcomes are labeled as Y_1 , Y_2 , Y_3 , Y_4 , and Y_5 , respectively, with Y_5 (proportion of time in care in 250 days) the primary outcome, and Y_1 through Y_4 the surrogate outcomes. We consider two covariates in the conditioning set of heterogeneous treatment effect: (a) initial Phase 1 treatment arm: SMS, CCT, or standard of care (SOC) and (b) time from Phase 1 enrollment to initial lapse in care. We used data from 215 participants who were assigned either Navigator or SMS+CCT in Phase 2 after a lapse in care in Phase 1 and fit a generalized linear model to obtain the counterfactual outcomes for simulation. The CATE functions for the surrogate outcomes and the final outcome are presented in Figure S10 (see Appendix I with more details). We implement a series of CARA designs as outlined in Section 5, including a simple sequential RCT, five adaptive designs that use a fixed choice from Y_1 through Y_5 , and the proposed TMLE-OSLAD adaptive design that evaluates and selects the best surrogate in real time. We also compared their performance with a naive policy that employs the latest available outcome at each time point to adapt treatment randomization probabilities. In each simulation, participants arrive following the enrollment schedule in the original trial with their observed covariates, and treatment assignments are generated sequentially according to the type of the simulated design and the covariate, treatment and outcome data of previously enrolled participants, then the outcomes are generated in response to the assigned treatment. This construction ensures that each observation is conditionally independent given the past. We simulate each design 500 times to assess the performance of TMLEs and different adaptive designs.

Results.

We compare regret over time obtained from implementing different design strategies (Figure S11 in Appendix I). The adaptive design using the earliest outcome Y_1 outperforms those based on later outcomes or the naive policy from $t = 2$ through $t = 11$. After $t = 12$, adaptive designs using later outcomes start achieving smaller regret for new subjects than the earliest outcome, as the underlying true CATE functions of later surrogates are closer to that for the final outcome and are estimated more precisely as data accumulate. However, this delayed improvement does not compensate the overarching benefit provided by the earliest surrogate, which delivers prompt benefits to participants by adopting surrogate-guided optimal personalized treatments learned from earlier and more abundant data. For the naive policy, although it initially matches the best performing adaptive design based on Y_1 , its aggressive shift to the most recent outcomes leads it to behave similarly to the suboptimal later-outcome designs, resulting in higher regret. In contrast, TMLE-OSLAD predominantly selects the earliest surrogates (Y_1 and Y_2) over time (Figure 3a). It yields a higher expected final outcome (Figure 3b) than designs that adhere to later outcomes without ongoing evaluation and approaches the performance of the oracle design using the earliest surrogate. TMLE results for the proposed estimands during the experiment and for additional estimands after the experiment are reported in Appendix I, showing robust performance.

7 Conclusions

In summary, this work provides a general TMLE-based Online Super Learner Adaptive Design Framework (TMLE-OSLAD) for constructing a real-time, data-driven ensemble system for adaptive experimental designs and shows how this framework contributes to the statistical theory and application of surrogate outcomes in adaptive designs. We propose a novel causal estimand for surrogate utility in sequential adaptive designs: the mean primary outcome under a counterfactual CARA design where, at enrollment, each participant's randomization is shifted toward the optimal individualized rule learned from a candidate surrogate. It captures both how well the surrogate guides optimal treatment assignment and the extent to which it enables earlier benefit for more participants, with the goal of improving participants' primary outcomes. Building on this, we define an Online Super Learner adaptive design that incorporates the proposed estimand to evaluate a library of candidate surrogates, selects the most useful surrogate based on TMLE inference, and dynamically adapts randomization probabilities by the selected surrogate-guided design.

This work on surrogate outcomes is a concrete application of our highly general TMLE-OSLAD framework, which provides statistical inference on the mean counterfactual outcome under each candidate adaptive design in a user-supplied set, and adaptively evaluates and updates the best-performing design over time to adapt treatment randomization for new participants. This unified framework enables principled evaluation of the opportunities and costs of diverse adaptive

designs varying in their design choices, quantifies how accumulating information benefits participants over time, and supports dynamic navigation of trade-offs among multiple designs during an ongoing experiment.

Methodologically, this framework advances non-parametric statistical inference for adaptive experiments: the proposed TMLE approach yields consistent, asymptotically normal estimators for the proposed estimand and other classical causal estimands, leveraging martingale theories, equicontinuity results, and Highly Adaptive Lasso. It broadly applies to estimation of various estimands in sequentially dependent settings like adaptive designs and time series.

Beyond the proposed context-specific estimand, further study of the computationally intensive “ideal estimand”, the expected outcome under the full data distribution induced by a candidate design (as is discussed in Appendix A.2), could provide more insight into the strengths and limitations of different estimands, as well as trade-offs between statistical properties and computational efficiency. We leave this to future work.

Disclosure Statement. The authors report there are no competing interests to declare.

References

- Atkinson, A. C., Biswas, A. and Pronzato, L. (2011), Covariate-balanced response-adaptive designs for clinical trials with continuous responses that target allocation probabilities, Technical Report NI11042-DAE, Isaac Newton Institute for Mathematical Sciences, Cambridge, UK.
- Bibaut, A., Dimakopoulou, M., Kallus, N., Chambaz, A. and van der Laan, M. (2021), ‘Post-contextual-bandit inference’, *Advances in Neural Information Processing Systems* **34**, 28548–28559.
- Brown, B. M. (1971), ‘Martingale central limit theorems’, *The Annals of Mathematical Statistics* **42**, 59–66.
- Bubeck, S., Cesa-Bianchi, N. et al. (2012), ‘Regret analysis of stochastic and nonstochastic multi-armed bandit problems’, *Foundations and Trends in Machine Learning* **5**(1), 1–122.
- Buyse, M., Molenberghs, G., Burzykowski, T., Renard, D. and Geys, H. (2000), ‘The validation of surrogate endpoints in meta-analyses of randomized experiments’, *Biostatistics* **1**(1), 49–67.
- Chambaz, A. and van der Laan, M. J. (2014), ‘Inference in targeted group-sequential covariate-adjusted randomized clinical trials’, *Scandinavian Journal of Statistics* **41**(1), 104–140.
- Chambaz, A., Zheng, W. and van der Laan, M. J. (2017), ‘Targeted sequential design for targeted learning inference of the optimal treatment rule and its mean reward’, *The Annals of Statistics* **45**(6), 2537–2564.
- Cheung, S. H., Zhang, L.-X., Hu, F. and Chan, W. S. (2014), ‘Covariate-adjusted response-adaptive designs for generalized linear models’, *Journal of Statistical Planning and Inference* **149**, 152–161.
- Chow, S.-C. and Chang, M. (2008), ‘Adaptive design methods in clinical trials—a review’, *Orphanet Journal of Rare Diseases* **3**(1), 11.
- Daniels, M. J. and Hughes, M. D. (1997), ‘Meta-analysis for the evaluation of potential surrogate markers’, *Statistics in Medicine* **16**(17), 1965–1982.
- Duan, W., Ba, S. and Zhang, C. (2021), Online experimentation with surrogate metrics: Guidelines and a case study, in ‘Proceedings of the 14th ACM International Conference on Web Search and Data Mining’, pp. 193–201.
- Elliott, M. R. (2023), ‘Surrogate endpoints in clinical trials’, *Annual Review of Statistics and its Application* **10**, 75–96.
- Freedman, L. S., Graubard, B. I. and Schatzkin, A. (1992), ‘Statistical validation of intermediate endpoints for chronic diseases’, *Statistics in Medicine* **11**(2), 167–178.

Gao, J., Hu, F. and Ma, W. (2024), 'Response-adaptive randomization procedure in clinical trials with surrogate endpoints', *Statistics in Medicine* **43**(30), 5911–5921.

Geng, E. H., Odeny, T. A., Montoya, L. M., Iguna, S., Kulzer, J. L., Adhiambo, H. F., Eshun-Wilson, I., Akama, E., Nyandieka, E., Guzé, M. A., Shade, S., Packel, L., Fox, B., Camlin, C., Thirumurthy, H., Lyons, C., Bukusi, E. A. and Petersen, M. L. (2023), 'Adaptive strategies for retention in care among persons living with hiv', *NEJM Evidence* **2**(4), EVIDoA2200076.

Gilbert, P. B. and Hudgens, M. G. (2008), 'Evaluating candidate principal surrogate endpoints', *Biometrics* **64**(4), 1146–1154.

Gill, R. D., van der Laan, M. J. and Wellner, J. A. (1995), 'Inefficient estimators of the bivariate survival function for three models', *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* **31**(3), 545–597.

Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S. and Athey, S. (2021), 'Confidence intervals for policy evaluation in adaptive experiments', *Proceedings of the National Academy of Sciences* **118**(15), e2014602118.

Hsu, J. Y., Kennedy, E. H., Roy, J. A., Stephens-Shields, A. J., Small, D. S. and Joffe, M. M. (2015), 'Surrogate markers for time-varying treatments and outcomes', *Clinical Trials* **12**(4), 309–316.

Hu, F. and Rosenberger, W. F. (2006), *The Theory of Response-Adaptive Randomization in Clinical Trials*, John Wiley & Sons.

Hu, J., Zhu, H. and Hu, F. (2015), 'A unified family of covariate-adjusted response-adaptive designs based on efficiency and ethics', *Journal of the American Statistical Association* **110**(509), 357–367.

Huang, X., Ning, J., Li, Y., Estey, E., Issa, J.-P. and Berry, D. A. (2009), 'Using short-term response information to facilitate adaptive randomization for survival clinical trials', *Statistics in Medicine* **28**(12), 1680–1689.

Larsen, N., Stallrich, J., Sengupta, S., Deng, A., Kohavi, R. and Stevens, N. T. (2024), 'Statistical challenges in online controlled experiments: A review of a/b testing methodology', *The American Statistician* **78**(2), 135–149.

Lattimore, T. and Szepesvári, C. (2020), *Bandit Algorithms*, Cambridge University Press.

Lin, D., Fleming, T. and De Gruttola, V. (1997), 'Estimating the proportion of treatment effect explained by a surrogate marker', *Statistics in Medicine* **16**(13), 1515–1527.

Luedtke, A. R. and van der Laan, M. J. (2016a), 'Statistical inference for the mean outcome under a possibly non-unique optimal treatment strategy', *The Annals of Statistics* **44**(2), 713–742.

- Luedtke, A. R. and van der Laan, M. J. (2016b), ‘Super-learning of an optimal dynamic treatment rule’, *The International Journal of Biostatistics* **12**(1), 305–332.
- Malenica, I., Bibaut, A. and van der Laan, M. J. (2021), ‘Adaptive sequential design for a single time-series’, *arXiv preprint arXiv:2102.00102* .
- Malenica, I., Coyle, J. R., van der Laan, M. J. and Petersen, M. L. (2024), ‘Adaptive sequential surveillance with network and temporal dependence’, *Biometrics* **80**(1), ujad007.
- McDonald, T. M., Maystre, L., Lalmas, M., Russo, D. and Ciosek, K. (2023), Impatient bandits: Optimizing recommendations for the long-term without delay, in ‘Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining’, pp. 1687–1697.
- Montoya, L. M., van der Laan, M. J., Luedtke, A. R., Skeem, J. L., Coyle, J. R. and Petersen, M. L. (2021), ‘The optimal dynamic treatment rule superlearner: considerations, performance, and application to criminal justice interventions’, *The International Journal of Biostatistics* **19**(1), 217–238.
- Murphy, S. A. (2003), ‘Optimal dynamic treatment regimes’, *Journal of the Royal Statistical Society Series B: Statistical Methodology* **65**(2), 331–355.
- Nowacki, A. S., Zhao, W. and Palesch, Y. Y. (2017), ‘A surrogate-primary replacement algorithm for response-adaptive randomization in stroke clinical trials’, *Statistical Methods in Medical Research* **26**(3), 1078–1092.
- Pearl, J. (2000), *Causality: Models, Reasoning, and Inference*, Cambridge University Press.
- Prentice, R. L. (1989), ‘Surrogate endpoints in clinical trials: definition and operational criteria’, *Statistics in Medicine* **8**(4), 431–440.
- Richardson, L., Zito, A., Greaves, D. and Soriano, J. (2023), ‘Pareto optimal proxy metrics’, *arXiv preprint arXiv:2307.01000* .
- Robbins, H. (1952), ‘Some aspects of the sequential design of experiments’, *Bulletin of the American Mathematical Society* **58**(5), 527–535.
- Robertson, D. S., Lee, K. M., López-Kolkovska, B. C. and Villar, S. S. (2023), ‘Response-adaptive randomization in clinical trials: From myths to practical considerations’, *Statistical Science* **38**(2), 185–208.
- Robins, J. M. (2004), Optimal structural nested models for optimal sequential decisions, in ‘Proceedings of the Second Seattle Symposium in Biostatistics’, Springer, pp. 189–326.
- Robins, J. M. and Greenland, S. (1992), ‘Identifiability and exchangeability for direct and indirect effects’, *Epidemiology* **3**(2), 143–155.

Rosenberger, W. F. and Sverdlov, O. (2008), ‘Handling covariates in the design of clinical trials’, *Statistical Science* **23**(3), 404–419.

Rosenberger, W. F., Vidyashankar, A. and Agarwal, D. K. (2001), ‘Covariate-adjusted response-adaptive designs for binary response’, *Journal of Biopharmaceutical Statistics* **11**(4), 227–236.

Simchi-Levi, D. and Wang, C. (2023), Multi-armed bandit experimental design: Online decision-making and adaptive inference, in ‘International Conference on Artificial Intelligence and Statistics’, PMLR, pp. 3086–3097.

Tamura, R. N., Faries, D. E., Andersen, J. S. and Heiligenstein, J. H. (1994), ‘A case study of an adaptive clinical trial in the treatment of out-patients with depressive disorder’, *Journal of the American Statistical Association* **89**(427), 768–776.

Thompson, W. R. (1933), ‘On the likelihood that one unknown probability exceeds another in view of the evidence of two samples’, *Biometrika* **25**(3-4), 285–294.

van der Laan, M. J. (2006), ‘Statistical inference for variable importance’, *The International Journal of Biostatistics* **2**(1).

van der Laan, M. J. (2008), ‘The construction and analysis of adaptive group sequential designs’, *U.C. Berkeley Division of Biostatistics. Working Paper 232*.
<https://biostats.bepress.com/ucbbiostat/paper232>.

van der Laan, M. J. (2023), ‘Higher order spline highly adaptive lasso estimators of functional parameters: Pointwise asymptotic normality and uniform convergence rates’, *arXiv preprint arXiv:2301.13354*.

van der Laan, M. J. and Luedtke, A. R. (2015), ‘Targeted learning of the mean outcome under an optimal dynamic treatment rule’, *Journal of Causal Inference* **3**(1), 61–95.

van der Laan, M. J. and Malenica, I. (2018), ‘Robust estimation of data-dependent causal effects based on observing a single time-series’, *arXiv preprint arXiv:1809.00734*.

van der Laan, M. J., Polley, E. C. and Hubbard, A. E. (2007), ‘Super learner’, *Statistical Applications in Genetics and Molecular Biology* **6**(1).

van der Laan, M. J. and Rose, S. (2011), *Targeted Learning: Causal Inference for Observational and Experimental Data*, Springer.

van der Laan, M. J. and Rose, S. (2018), *Targeted Learning in Data Science*, Springer.

van der Laan, M. J. and Rubin, D. (2006), ‘Targeted maximum likelihood learning’, *The International Journal of Biostatistics* **2**(1).

VanderWeele, T. J. (2013), ‘Surrogate measures and consistent surrogates’, *Biometrics* **69**(3), 561–565.

Weir, C. J. and Taylor, R. S. (2022), ‘Informed decision-making: Statistical methodology for surrogacy evaluation and its role in licensing and reimbursement assessments’, *Pharmaceutical Statistics* **21**(4), 740–756.

Yang, J., Eckles, D., Dhillon, P. and Aral, S. (2024), ‘Targeting for long-term outcomes’, *Management Science* **70**(6), 3841–3855.

Zhan, R., Hadad, V., Hirshberg, D. A. and Athey, S. (2021), Off-policy evaluation via adaptive weighting with data from contextual bandits, in ‘Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining’, pp. 2125–2135.

Zhang, K., Janson, L. and Murphy, S. (2021), ‘Statistical inference with m-estimators on adaptively collected data’, *Advances in Neural Information Processing Systems* **34**, 7460–7471.

Zhang, L.-X., Hu, F., Cheung, S. H. and Chan, W. S. (2007), ‘Asymptotic properties of covariate-adjusted response-adaptive designs’, *The Annals of Statistics* **35**(3), 1166–1182.

Zhao, W., Ma, W., Wang, F. and Hu, F. (2022), ‘Incorporating covariates information in adaptive clinical trials for precision medicine’, *Pharmaceutical Statistics* **21**(1), 176–195.

Zheng, W. and van der Laan, M. J. (2011), *Cross-Validated Targeted Minimum-Loss-Based Estimation*, Springer New York, New York, NY, pp. 459–474.

Zhu, H. (2015), ‘Covariate-adjusted response adaptive designs incorporating covariates with and without treatment interactions’, *Canadian Journal of Statistics* **43**(4), 534–553.

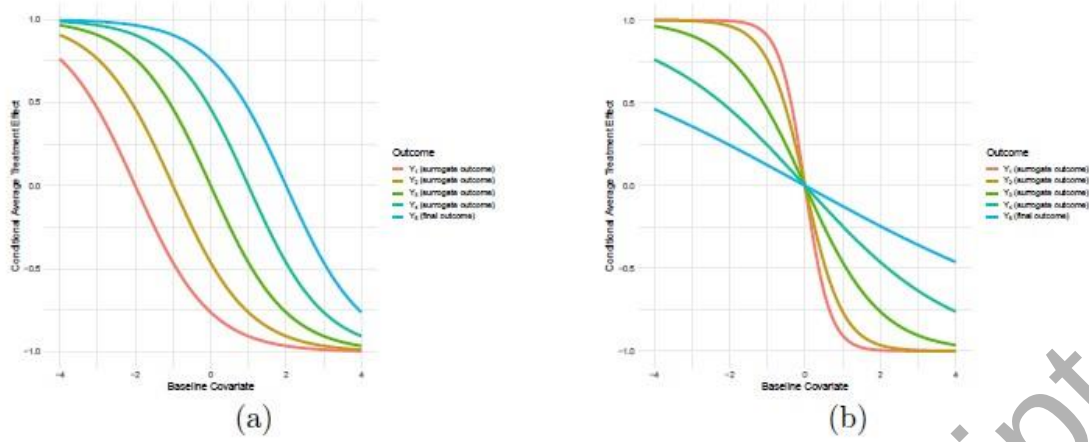


Figure 1: Conditional Average Treatment Effect (CATE) functions for the simulation scenarios: (a) Scenario 1, where later surrogates have higher utility; (b) Scenario 2, where earlier surrogates have higher utility. Each curve depicts the CATE function for a candidate outcome, Y_1 through Y_5 .

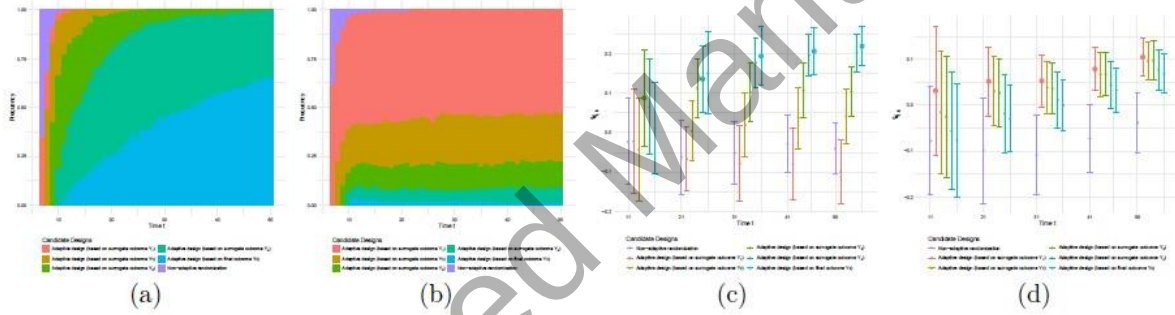


Figure 2: Frequency of design selection by TMLE-OSLAD at each time point. Panels (a) and (b) correspond to Scenario 1 and Scenario 2, respectively. Each color represents one candidate design, including adaptive designs based on outcomes Y_k ($k \in \{1, \dots, 5\}$) and the non-adaptive design. The height of each bar indicates the proportion of simulation runs in which that design attains the highest lower bound of the 95% confidence interval for $\hat{\psi}_{t,k}$ among all candidate designs at time t , reflecting the selection mechanism of TMLE-OSLAD. Panels (c) and (d) display the selection mechanism at illustrative time points ($t = 11, 21, 31, 41, 50$) in a single simulation run under Scenarios 1 and 2, respectively. Colored points show the TMLE point estimates of $\psi_{t,k}$ (y-axis) for evaluating each candidate design at each time point (x-axis). Vertical bars indicate the corresponding 95% Wald confidence intervals. At each time point, the selected design (defined as the one with the largest lower confidence bound among all candidate designs) is highlighted by a circular outline.

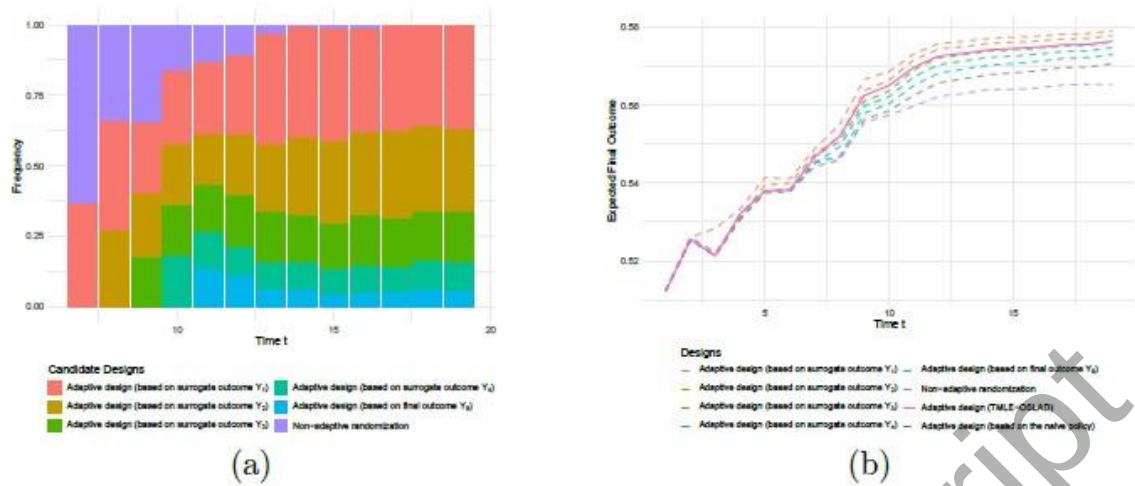


Figure 3: Panel (a) shows the frequency of candidate adaptive design selections by TMLE-OSLAD at each time point in the ADAPT-R trial simulation. Panel (b) shows the cumulative average expected final outcome over time obtained from implementing different designs.

Table 1: True target estimand $\Psi_{t,k}$ had participants enrolled from $1, \dots, t-K$ were adaptively treated by a candidate surrogate Y_k in Scenarios 1 and 2 ($t = 11, 21, 31, 41, 50$). Expected outcomes under oracle surrogate are shown in bold.

(a) Scenario 1						
t	$\Psi_{t,RCT}$	$\Psi_{t,1}$	$\Psi_{t,2}$	$\Psi_{t,3}$	$\Psi_{t,4}$	$\Psi_{t,5}$
11	0.000	-0.016	0.053	0.078	0.066	0.033
21	0.000	-0.008	0.083	0.145	0.177	0.173
31	0.000	-0.005	0.091	0.161	0.203	0.208
41	0.000	-0.002	0.094	0.169	0.215	0.225
50	0.000	-0.001	0.096	0.173	0.222	0.234
(b) Scenario 2						
t	$\Psi_{t,RCT}$	$\Psi_{t,1}$	$\Psi_{t,2}$	$\Psi_{t,3}$	$\Psi_{t,4}$	$\Psi_{t,5}$
11	0.000	0.073	0.057	0.039	0.019	0.004
21	0.000	0.087	0.081	0.073	0.061	0.038
31	0.000	0.090	0.086	0.081	0.073	0.052
41	0.000	0.091	0.089	0.085	0.078	0.060
50	0.000	0.092	0.090	0.087	0.081	0.065

Table 2: Performance of TMLEs for other causal estimands at the end of experiments in Scenarios 1 and 2 across Monte Carlo simulations.

(a) Scenario 1					
Estimand	Notation	Truth	Bias ($\times 10^{-3}$)	Var ($\times 10^{-3}$)	Coverage (%)
Average treatment effect on the final outcome Y_5	$\psi_{T_{end}}^{ATE}$	0.468	3.94	3.92	95.8
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_1	$\psi_{T_{end}}^{d_{n,1}^*}$	0.011	-1.01	2.85	94.4
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_2	$\psi_{T_{end}}^{d_{n,2}^*}$	0.130	-0.87	2.02	93.8
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_3	$\psi_{T_{end}}^{d_{n,3}^*}$	0.235	-0.08	1.62	93.6
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_4	$\psi_{T_{end}}^{d_{n,4}^*}$	0.312	-1.01	0.89	94.6
Expected final outcome Y_5 under the estimated ODTR based on Y_5	$\psi_{T_{end}}^{d_{n,5}^*}$	0.340	0.42	0.72	95.6
(b) Scenario 2					
Estimand	Notation	Truth	Bias ($\times 10^{-3}$)	Var ($\times 10^{-3}$)	Coverage (%)
Average treatment effect on the final outcome Y_5	$\psi_{T_{end}}^{ATE}$	0.000	-1.17	3.99	94.8
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_1	$\psi_{T_{end}}^{d_{n,1}^*}$	0.120	-0.25	0.57	93.8
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_2	$\psi_{T_{end}}^{d_{n,2}^*}$	0.120	-0.32	0.59	94.2
Expected final outcome Y_5 under the estimated	$\psi_{T_{end}}^{d_{n,3}^*}$	0.120	-0.57	0.61	93.6

(a) Scenario 1					
Estimand	Notation	Truth	Bias ($\times 10^{-3}$)	Var ($\times 10^{-3}$)	Coverage (%)
ODTR based on surrogate Y_3					
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_4	$\psi_{T_{end}}^{d_{n,4}^*}$	0.119	-0.38	0.60	94.6
Expected final outcome Y_5 under the estimated ODTR based on Y_5	$\psi_{T_{end}}^{d_{n,5}^*}$	0.114	0.31	0.81	93.0

Table 3: Performance of TMLEs for other causal estimands at the end of experiments in Scenarios 1 and 2 across Monte Carlo simulations.

(a) Scenario 1					
Estimand	Notation	Truth	Bias ($\times 10^{-3}$)	Var ($\times 10^{-3}$)	Coverage (%)
Average treatment effect on the final outcome Y_5	$\psi_{T_{end}}^{ATE}$	0.468	3.94	3.92	95.8
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_1	$\psi_{T_{end}}^{d_{n,1}^*}$	0.011	-1.01	2.85	94.4
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_2	$\psi_{T_{end}}^{d_{n,2}^*}$	0.130	-0.87	2.02	93.8
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_3	$\psi_{T_{end}}^{d_{n,3}^*}$	0.235	-0.08	1.62	93.6
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_4	$\psi_{T_{end}}^{d_{n,4}^*}$	0.312	-1.01	0.89	94.6
Expected final outcome Y_5 under the estimated ODTR based on Y_5	$\psi_{T_{end}}^{d_{n,5}^*}$	0.340	0.42	0.72	95.6
(b) Scenario 2					
Estimand	Notation	Truth	Bias ($\times 10^{-3}$)	Var ($\times 10^{-3}$)	Coverage (%)
Average treatment effect on the final outcome Y_5	$\psi_{T_{end}}^{ATE}$	0.000	-1.17	3.99	94.8
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_1	$\psi_{T_{end}}^{d_{n,1}^*}$	0.120	-0.25	0.57	93.8
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_2	$\psi_{T_{end}}^{d_{n,2}^*}$	0.120	-0.32	0.59	94.2
Expected final outcome Y_5 under the estimated	$\psi_{T_{end}}^{d_{n,3}^*}$	0.120	-0.57	0.61	93.6

(a) Scenario 1					
Estimand	Notation	Truth	Bias ($\times 10^{-3}$)	Var ($\times 10^{-3}$)	Coverage (%)
ODTR based on surrogate Y_3					
Expected final outcome Y_5 under the estimated ODTR based on surrogate Y_4	$\psi_{T_{end}}^{d_{n,4}^*}$	0.119	-0.38	0.60	94.6
Expected final outcome Y_5 under the estimated ODTR based on Y_5	$\psi_{T_{end}}^{d_{n,5}^*}$	0.114	0.31	0.81	93.0